

The CTB diffraction model these masks hang on

All-reflective 2-DM coronagraph relay — masks applied to the complex field in MATLAB (no engine change)

Light path — compact deck (ctb_dcr.in, 31 elts)

source → OAP1 → **DM1** → ⟨p2p⟩ → DM2 → OAP2 → [focus] → OAP3 →
→ **apodizer** → OAP4 → **FPM** [focus] → OAP5 → **Lyot** →
→ OAP6 → [field stop] → OAP7 → OAP8 → ExitPupil → **FPA**
stations: DM1=2 DM2=5 Apodizer=13 FPM=17 Lyot=20 EP=30 FPA=31
— only DM1→DM2 p2p + the 3 through-focus quartets propagated; other gaps are geometric jumps.

Light path — full model (ctb_s2s_dcr.in, 44 elts)

source → OAP1 → ⟨p2p⟩ → DM1 → ⟨p2p⟩ → DM2 → ⟨p2p⟩ → OAP2 → [focus] →
→ OAP3 → ⟨p2p⟩ → apodizer → ⟨p2p⟩ → OAP4 → [FPM] → OAP5 → ⟨p2p⟩ → Lyot →
→ ⟨p2p⟩ → OAP6 → [field stop] → OAP7 → ⟨p2p⟩ → OAP8 → ExitPupil → **FPA**
stations: DM1=2 DM2=5 Apodizer=16 FPM=22 Lyot=27 EP=43 FPA=44
— EVERY inter-optic leg propagated (8 NFPlane p2p props added); adds the inter-optic diffraction the compact deck omits.

Mask stations Apodizer (pupil) · FPM (focus) · Lyot (pupil) — passive Reference markers; masks added in MATLAB.

How a mask is applied (stage-1 contract)

1. propagate to the plane → multiply the complex field by the mask array → continue.
 2. `macos.apodize` (real 0–1 amplitude) · `macos.apodize_complex` (complex).
 3. subsequent reads use `'reset_trace', false` to keep the mask.
- ✓ No source tilt, no reference re-alignment, no engine edit.

Deterministic sampling (engine geometry, not dx_at at NF planes)

focal pitch $dx_f = \lambda R / (N \cdot dx_{sph})$
FPM-local λ/D $\lambda R / D_{beam}$ (metres)
FPA λ/D $\lambda \cdot R_{fpa} / D_{ep} / dx_{FPA}$

On this bench (N=1024, 500 nm)

- λ/D at FPA = 4.03 px (N/pupil; zero-pad N → finer PSF)
- occulter centred on floor(N/2) = FFT DC pixel (the half-pixel rule)
- masks re-evaluated PER λ on each λ 's grid (chromatic study)
- 2-DM relay → full annular dark zone is the fair scoring region

Validation

FPM through-focus leg vs MATLAB PROPER: peak-norm corr 1.000000,
dx ratio 1.0000, 0.000 px centroid offset.

Literature coronagraph mask families

Standard mask/apodizer pairings on the existing apodize machinery — formulae verified verbatim from the papers

Five families — driver · result (3–15 λ/D , N=1024)

Band-limited Lyot `ctb_bandlimited`

null 4.3e5; $(1-\epsilon)R$ trim verified; 1.1× chromatic

Vortex + matched Lyot `ctb_vortex_matched`

3.2× throughput vs unmatched (star→ring)

APLC (prolate) `ctb_aplc`

139× deeper than BLC at equal 27% thru

Roddier π / dual-zone `ctb_phase_masks`

first literature complex focal masks

all six, head-to-head `ctb_mask_compare`

table + scatter + .mat

Mask builders (new, reusable)

`ctb_mask_bandlimited` · `ctb_apod_prolate` · `ctb_mask_phase`

Shared helpers (reused)

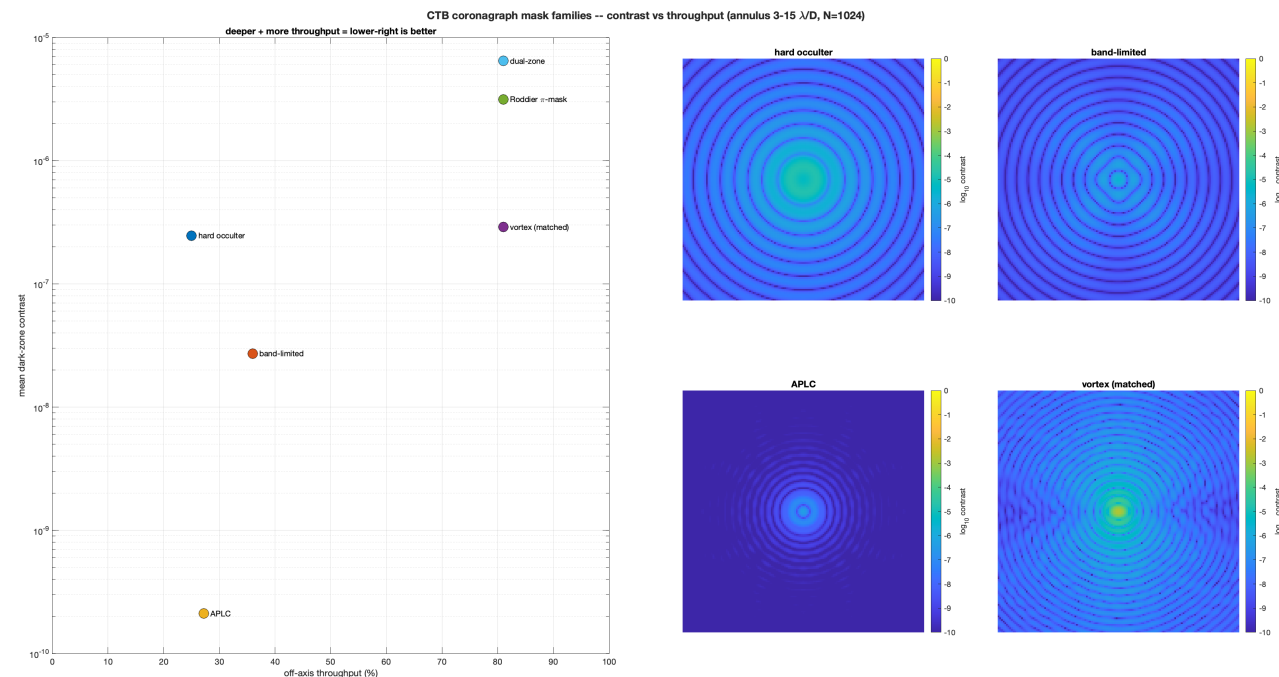
`ctb_mask_disk/softcircle` · `dark_zone_metrics` · `radial_contrast`

Saved (in the example dir)

one PNG per driver + `ctb_mask_compare.mat` (results table) + `CTB_PROP_STATUS.md`

User-changeable parameters

`model_size`, `inner/outer_lamD`, `rx/elt`; `order/form/epsilon` (BLC),
`charges/lyot_fracs` (vortex), `r_occ_lamD/r_lyot_frac` (APLC),
`rho0_lamD/dz_*` (phase)



HLC deferred to the FALCO integration — its FPM is a co-optimization product, not a formula.

References

Formulae extracted verbatim from the ar5iv/arXiv LaTeX of each source

Band-limited Lyot

- Kuchner & Traub 2002, ApJ 570, 900 — 4th-order 1-sinc amplitude mask (Eq. 7/8); Lyot trim ($1-\epsilon$). [astro-ph/0203455]
- Kuchner, Crepp & Ge 2005, ApJ 628, 466 — 8th-order masks (Eq. 12/13, $m=1, l=3$). [astro-ph/0411077]

APLC (apodized-pupil Lyot)

- Soummer 2005, ApJ 618, L161 — prolate apodizer eigenvalue eqn (Eq. 3). [astro-ph/0412221]
- Soummer, Aime & Falloon 2003, A&A 397, 1161 · Soummer et al. 2011, ApJ 729, 144 (GPI config, $2.8 \lambda/D$).

Phase masks

- Roddier & Roddier 1997, PASP 109, 815 — π spot, radius $0.53 \lambda/D$, no apodizer (flux balance).
- N'Diaye et al. 2012, A&A 538, A55 — achromatic dual-zone (pure-phase, wavelength-sliding). [arXiv:1111.3194]
- Soummer, Dohlen & Aime 2003, A&A 403, 369 — dual-zone phase mask.

Vortex

- Mawet et al. 2005, ApJ 633, 1191 · Foo, Palacios & Swartzlander 2005, Opt. Lett. 30, 3308.
- Jenkins 2008, MNRAS 384, 515 — ideal even-charge field = 0 inside pupil (Eq. 1). [arXiv:0709.0153]

Note: a WebFetch summarizer hallucinated equations on every topic; all forms above were reconciled against the raw paper HTML.